

Mini Cases

Risk Management

2025/2026

Please upload your solutions and indicate which Mini Cases (MCs) you have solved before **June 17, 2026 at 7am**. To do so, go to the assignment section on Canvas and choose “Unit 1 Indication and Submission of MCs”. Then answer the true/false questions (Questions 1 to 6) and upload your solutions file (Question 7). **There will be no deadline extensions! Only complete submissions containing answers to the T/F questions and a solutions upload will be considered, your last complete attempt counts!** You can either write your solutions using software or scan your handwritten solutions. Please note that you may only submit one solutions file. If your solutions consists of more than one file (e.g. 1 pdf and 1 Excel file), please upload a .zip file. Uploaded files will be checked for plagiarism. The solutions in your file should be self-explanatory - not only for the instructor, but for other students as well. Indicate only Mini Cases you feel confident to present.

Important remark for the presentations: Make sure to provide a concise explanation of your solutions. In particular, when applying any formula from the lecture materials or other sources, you may be asked to explain the structure of your solutions and provide the intuition and underlying concepts behind the formula used.

Mini Case 4.7.

Consider a payer swap with a fixed rate of 0.06 (in decimals) and a maturity of 3 years. The fixed payments are paid annually and the floating leg pays the 3M Euribor. Assuming a flat interest rate of 0.05 (in decimals), what is the modified duration of the swap?

Mini Case 4.8.

Consider a firm with debt D equal to 60 and equity E equal to 30. The modified durations of assets MD_A and debt MD_D are 16 and 12, respectively.

- What is the modified duration of equity MD_E ?
- Next, assume that the firm wants to hedge the entire risk exposure by entering an IRS with $MD_{IRS,assets} \approx 0.5$ and $MD_{IRS,debt} = 12.5$. Compute the approximately correct notional N to achieve the hedging goal.
- Does the firm enter a payer or a receiver swap?
- Finally, assume that after the firm has entered into the swap, interest rates increase by 5 percentage points. Show that the firm doesn't make a profit or suffer a loss.

Mini Case 4.9.

Consider a firm with debt D of 50 and equity E of 50. The modified duration of debt MD_D is equal to 14. To achieve a perfect hedge, the firm entered into a swap where the floating leg has a modified duration of 0.25 and the fixed leg's duration is 25.

- Assuming that you know that the modified duration of (unhedged) equity MD_E is negative, did the firm enter a payer or receiver swap? What are the values for $MD_{IRS,assets}$ and $MD_{IRS,debt}$?

- Given that the notional of the swap is equal to 40, what is the modified duration of assets MD_A ?
- Finally, what is the modified duration of (unhedged) equity?

Mini Case 5.1.

Consider an Austrian firm with operations in Austria A and Poland P . Assume that the exchange rate X is quoted in EUR per PLN. You will find in the provided data files three time series A , P and X that each have normally distributed percentage changes, e.g. $\frac{\Delta A}{A} \sim N(\mu_A, \sigma_A)$.

- Calculate σ_A , σ_P , σ_X , $\text{cor}(A, P)$, $\text{cor}(A, X)$ and $\text{cor}(P, X)$.
- Calculate a 1% VaR for the entire firm assuming a delta-normal model. Use as values for A , P and X the values as of time period 1.
- What is a good estimate for the "marginal" VaR of FX risk?

Mini Case 5.2.

Consider an oil producing company in Saudi Arabia. The revenues of the company accrue from selling oil, which is traded in US Dollar. The company pays salaries and all other bills in local currency (SAR). It generates annual revenues of 500 USD and its total costs are 2000 SAR. Assume that the costs are constant in each period, and that revenues and cost accrue for infinite periods. The interest rate is 3%. The exchange rate is 8 SAR per 1 USD.

- What is the value of the operation in SAR?
- By how much does the PV change if the exchange rate increases by 30% from 8 to 10.4? Assume that the sensitivity of cash flows to the exchange rate is constant over time. Explain.

- What is the present value of the revenues in SAR, using the original exchange rate?
- Suppose now that the exchange rate has an annual standard deviation of 10%. Calculate the 1% VaR for the present value of revenues in SAR, using the delta normal method.
- Discuss the similarities and differences in the measures calculated above.

Mini Case 5.3.

Consider an Austrian firm with operations in the UK. To measure the associated exchange risk, the firm aggregates the cash flows into five buckets and uses one key maturity for each bucket. The buckets, key maturities, exchange rates, and the operations' future cash flows are given in "Mini_Case_5_3.xlsx" or "Mini_Case_5_3.Rdata". Assume that the GBP and EUR interest rates are zero and that today is 2021-01-01.

- What is the value of the operations if the firm hedges the exchange rate risk with forwards?
- What is the value of the operations if the firm doesn't hedge and uses today's spot rate for all cash flows?
- Assume that after 1 year the exchange rate unexpectedly decreases by 15% from 0.8697 to 0.7392. What happens to the value of the remaining cash flows in Question 2?
- Finally, assume that the firm wants to hedge itself against FX shocks after 1 year. How much EUR notional of a forward contract is required?